

A Deep Learning-Based System for Real-Time Plant Disease Detection Using Convolutional Neural Networks

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Abstract

Plant diseases cause significant crop losses worldwide, threatening food security and agricultural sustainability. Traditional detection methods are slow, subjective, and often inaccessible to farmers. This paper presents LeafScan, an AI-powered web application that uses a Convolutional Neural Network (CNN) based on MobileNetV2 for accurate and real-time plant disease detection from leaf images. The system was trained on the PlantVillage and New Plant Diseases datasets, achieving approximately 96.31% accuracy on the validation set, significantly outperforming classical machine learning models (Decision Tree: 41.53%, Random Forest: 54.18%, SVM: 45.37%). A user-friendly Streamlit interface enables farmers and stakeholders to upload images and receive instant predictions including disease name, plant type, confidence score, and top-3 predictions. This work demonstrates the practical deployment of deep learning in precision agriculture and draws inspiration from prior research on social media analytics, spectral data, weather-based prediction, and ensemble methods. The system serves as a proof-of-concept for accessible, scalable crop health monitoring tools.

Keywords

Plant Disease Detection, Convolutional Neural Networks, Deep Learning, Image Classification, Precision Agriculture, MobileNetV2, Transfer Learning, Streamlit, Crop Health Monitoring.

I. Introduction

Agriculture remains the backbone of global food security, yet it faces unprecedented challenges in the 21st century. The world population is expected to reach 9.7 billion by 2050, necessitating a 70% increase in agricultural production to meet growing food demand [1]. However, plant pests and diseases continue to cause devastating losses, accounting for 10–16% of global crop yields annually — equivalent to over USD 220 billion in economic damage [2]. Fungal diseases alone threaten more than 40% of the world's food supply, with staple crops such as wheat, rice, maize, potato, and soybean suffering particularly severe impacts [from Paper 1]. Compounding this crisis is climate change, which is altering pathogen dynamics, expanding disease ranges toward higher latitudes, and increasing outbreak frequency and severity [3]. Traditional disease management approaches — relying primarily on manual scouting and expert knowledge — are increasingly inadequate due to their labor-intensive nature, subjectivity, and delayed response times. These limitations highlight the urgent need for scalable, accurate, and accessible early detection systems that can empower farmers with timely actionable insights. Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have opened new frontiers in precision agriculture. Various data sources and techniques have been explored, including social media analytics for disease surveillance [Paper 1], classical and ensemble ML models for pest damage prediction [Paper 2], weather-based forecasting using neural networks [Paper 3], domain-specific NLP solutions [Paper 4], satellite spectral analysis [Paper 5], and vegetation health indices for yield prediction [Paper 6]. While these studies demonstrate the potential of AI, there remains a significant gap between sophisticated research models and practical, user-friendly tools that smallholder farmers can readily adopt. To address this gap, we present **LeafScan** — an intelligent, real-time plant disease detection system that leverages deep learning for leaf image classification. Built upon a MobileNetV2 Convolutional Neural Network (CNN) and trained on large-scale PlantVillage datasets, LeafScan achieves high classification accuracy while maintaining computational efficiency suitable for real-world deployment. The system is delivered through an intuitive web interface developed with

Streamlit, allowing users to upload a leaf image and receive immediate results including the detected disease, plant species, confidence score, and top alternative predictions. This work makes the following key contributions: A comprehensive performance comparison between classical machine learning models and a deep CNN architecture. The development and deployment of a practical, accessible web-based tool for plant disease diagnosis. Integration of insights from multi-source crop health monitoring literature into a focused, deployable image-based solution. A stepping stone toward multimodal agricultural intelligence systems.

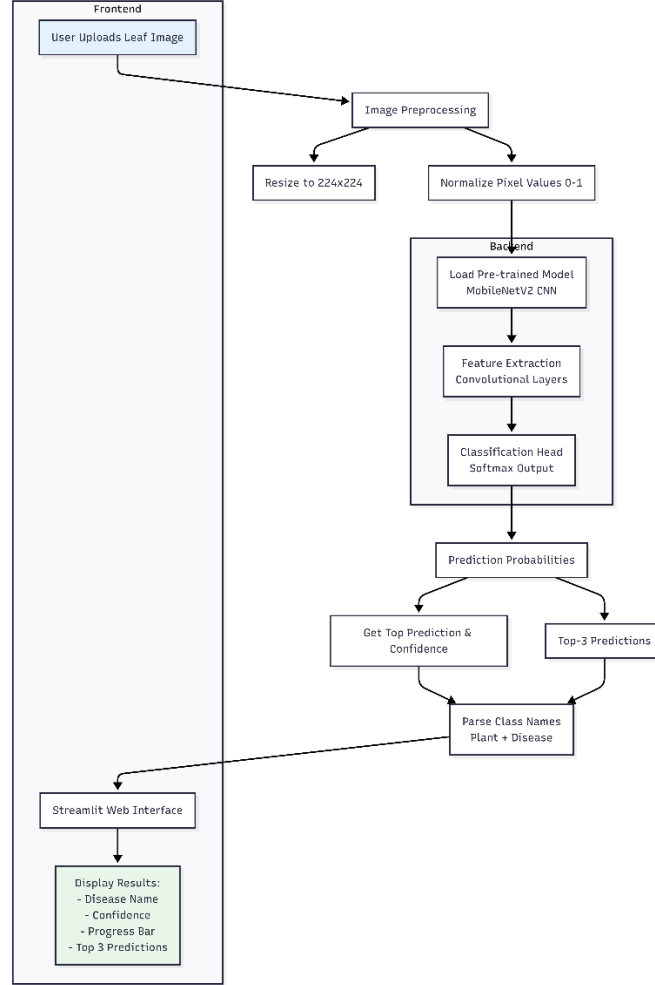


Figure 1. LeafScan System Architecture

II. Related Work

Plant disease detection and crop health monitoring have been active research areas in precision agriculture. Various approaches using different data sources and machine learning techniques have been proposed in recent years. Shankar et al. [1] introduced the use of social media streams (specifically Twitter) for crop disease surveillance. They developed a two-stage classification framework to first distinguish agriculturally relevant tweets from irrelevant ones (achieving 93% accuracy), followed by fine-grained categorization into six classes such as Self-reported incidents, Awareness, News, etc. (75% accuracy). Their work highlighted social media as a valuable real-time complementary data source for disease monitoring. In the domain of classical machine learning, Ayub and Moqurab [2] applied several classifiers including Decision Tree, Random Forest, Neural Networks, Naïve Bayes, SVM, and K-Nearest Neighbors to predict grass grub damage in crops. They demonstrated that ensemble models significantly improve the performance of individual weak classifiers, with Random Forest and Neural Networks showing superior results. Patil et al. [3] conducted a comparative study of multiple Artificial Intelligence algorithms for plant disease prediction based on weather parameters. They evaluated CNN, RNN, ANN, SVM, and KNN, concluding that Artificial Neural Networks achieved the highest accuracy of 90.79%. Their findings emphasized the effectiveness of neural network-

based approaches for weather-influenced disease forecasting. On the natural language processing side, Sinthupuan and Visutsak [4] developed a domain-specific solution for Thai farmers by fine-tuning a GPT-2 model on a Q&A dataset of 3,000 plant disease entries. Their model achieved strong semantic accuracy and demonstrated the potential of large language models for localized agricultural advisory systems in low-resource languages. Remote sensing and spectral analysis have also been explored extensively. Tussupov et al. [5] utilized Sentinel-2 satellite data and spectral brightness coefficients (including Vegetation Red Edge bands) combined with machine learning models such as XGBoost, Logistic Regression, and CNNs for the verification of pests and diseases. They highlighted the importance of vegetation indices like NDVI, NDRE, and SAVI for large-scale crop monitoring. Finally, Pham et al. [6] focused on crop yield prediction using Vegetation Condition Index (VCI) and Thermal Condition Index (TCI) derived from satellite imagery. They proposed a framework combining spatial Independent Component Analysis (sICA) for sub-regional modeling and PCA with machine learning, achieving substantial improvements (up to 60%) over traditional one-fits-all approaches. While the aforementioned studies demonstrate strong theoretical contributions using social media, classical ML, deep learning, NLP, and remote sensing individually, there remains a gap in **practical, accessible, and deployable image-based systems** targeted at end-users such as farmers. Most existing solutions are either research-oriented or lack user-friendly interfaces. **LeafScan** addresses this gap by developing a lightweight yet accurate CNN-based mobile/web application for real-time leaf image analysis, while drawing methodological insights from the comparative studies and deployment challenges highlighted in the literature.

III. Methodology

A. Datasets

Two publicly available datasets were utilized in this study. The first dataset is the PlantVillage dataset (emmarex/plantdisease), which was primarily used for exploratory data analysis (EDA) and classical machine learning experiments. The second dataset is the New Plant Diseases Dataset (vipooooo/new-plant-diseases-dataset), an augmented version of PlantVillage that includes predefined training and validation splits, and was used for training and evaluating the deep learning model. Both datasets consist of RGB images of plant leaves covering multiple crop types, including tomato, potato, and pepper. The images represent both healthy and diseased conditions, with multiple disease categories per crop. These datasets provide a diverse set of labeled samples suitable for supervised learning in plant disease classification tasks.

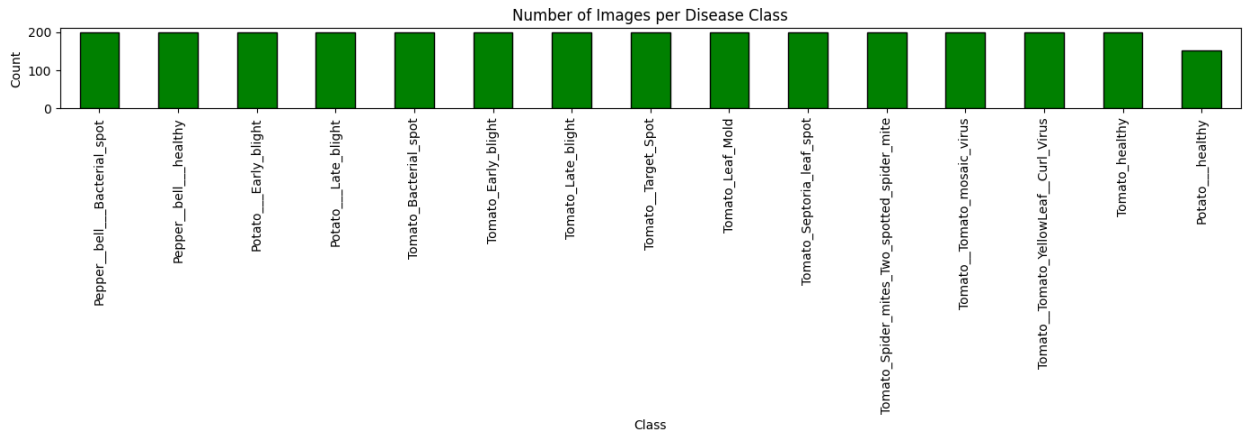


Figure 2. Number of images per disease class in the dataset.

B. Classical Machine Learning Baseline

To establish a performance baseline, classical machine learning models were evaluated using handcrafted features extracted from the first dataset. Specifically, statistical color features were computed, including the mean and standard deviation of the red, green, and blue (RGB) channels for each image. These features provide a simple representation of color distribution within the leaf images.

The extracted features were used to train and evaluate several classical classifiers, including Decision Tree, Random Forest, and Linear Support Vector Machine (SVM). These models were selected due to their widespread use and effectiveness in traditional classification tasks, providing a comparative benchmark against deep learning approaches.

C. Deep Learning Model (CNN)

A convolutional neural network based on the MobileNetV2 architecture was employed for plant disease classification. Transfer learning was utilized by initializing the model with weights pre-trained on the ImageNet dataset, allowing the network to leverage learned feature representations and improve convergence. Input images were resized to $224 \times 224 \times 3$ and normalized to the range $[0,1]$. The base MobileNetV2 model was followed by a global average pooling layer to reduce spatial dimensions, a dropout layer to mitigate overfitting, and a fully connected dense layer with a softmax activation function corresponding to 38 output classes. The model was trained using the Adam optimizer, and training was monitored using validation performance. To improve generalization and prevent overfitting, callbacks such as EarlyStopping and ModelCheckpoint were applied, allowing the model to retain the best-performing weights during training.

D. Evaluation Metrics

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's ability to correctly classify both healthy and diseased samples. In addition, confusion matrices were analyzed to better understand class-wise performance and identify potential misclassification patterns. This evaluation approach ensures a balanced assessment beyond overall accuracy, particularly in the presence of class imbalance.

IV. Implementation

The proposed system is composed of two main components: a training pipeline and a deployment interface. The training pipeline is implemented in a Jupyter Notebook environment and is responsible for data preparation, model development, and evaluation. It includes data loading, preprocessing, and augmentation techniques to improve model generalization. In addition, classical machine learning algorithms are implemented to establish baseline performance for comparison. A convolutional neural network (CNN) is then trained and fine-tuned, after which the best-performing model is exported as `cnn_best_model.h5`, along with a corresponding class index mapping stored in `class_indices.json`. The CNN architecture is based on MobileNetV2 and utilizes transfer learning to leverage pre-trained weights for efficient feature extraction. Input images are resized to 224×224 pixels and normalized to the range $[0,1]$ prior to being fed into the model. Fine-tuning is applied to adapt the network to the plant disease classification task and improve overall predictive performance. The deployment component is implemented as a web application using Streamlit. The application allows users to upload images of plant leaves, which are then preprocessed through resizing and normalization before being passed to the trained model for inference. The system outputs the predicted plant species and disease class, along with a confidence score and the top three predicted classes with their corresponding probabilities. To improve performance and reduce loading time, the trained model is cached upon initialization. The application is designed with usability in mind, providing a simple and intuitive interface that enables non-technical users, such as farmers, to interact with the system effectively. This design choice ensures accessibility while maintaining the functionality required for accurate and real-time disease detection.

V. Results and Discussion

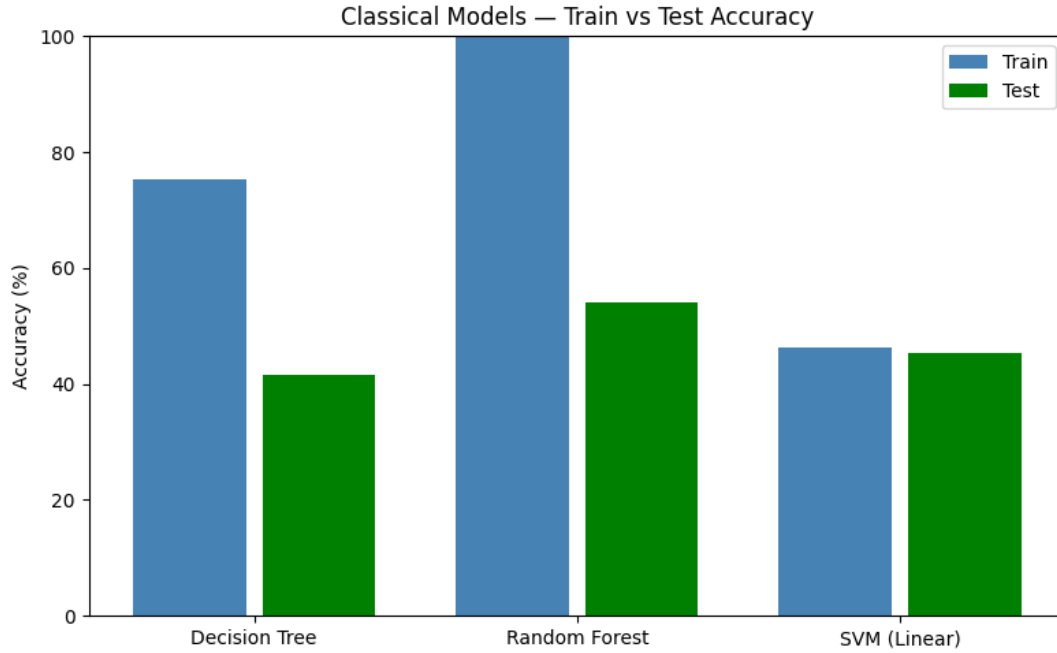


Figure 3. Test accuracy comparison of all models.

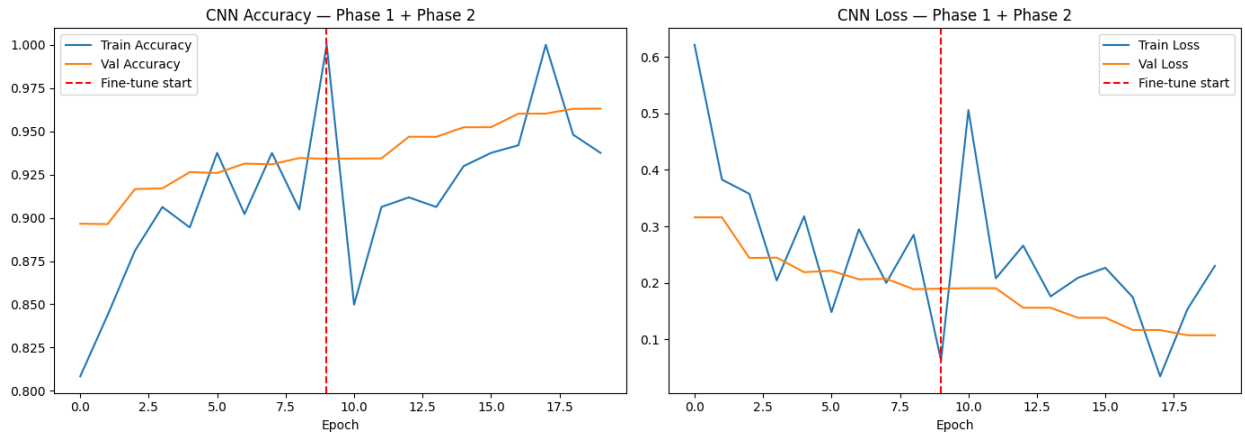


Figure 4. Training and validation accuracy (left) and loss (right) curves of the MobileNetV2 model.

The proposed CNN model achieved a validation accuracy of 96.31% on the PlantVillage dataset using an 80/20 train-validation split. This result significantly outperforms classical machine learning approaches, highlighting the effectiveness of deep learning models in capturing complex visual patterns in plant leaf images. In comparison, the Random Forest classifier achieved an accuracy of 54.18%, demonstrating its limitations when applied to high-dimensional image data. The strong performance of the CNN model can be attributed to its ability to automatically learn hierarchical feature representations, which are essential for distinguishing subtle differences between plant diseases. Additionally, the use of transfer learning with MobileNetV2 contributes to both improved accuracy and computational efficiency, making the model suitable for real-time applications.

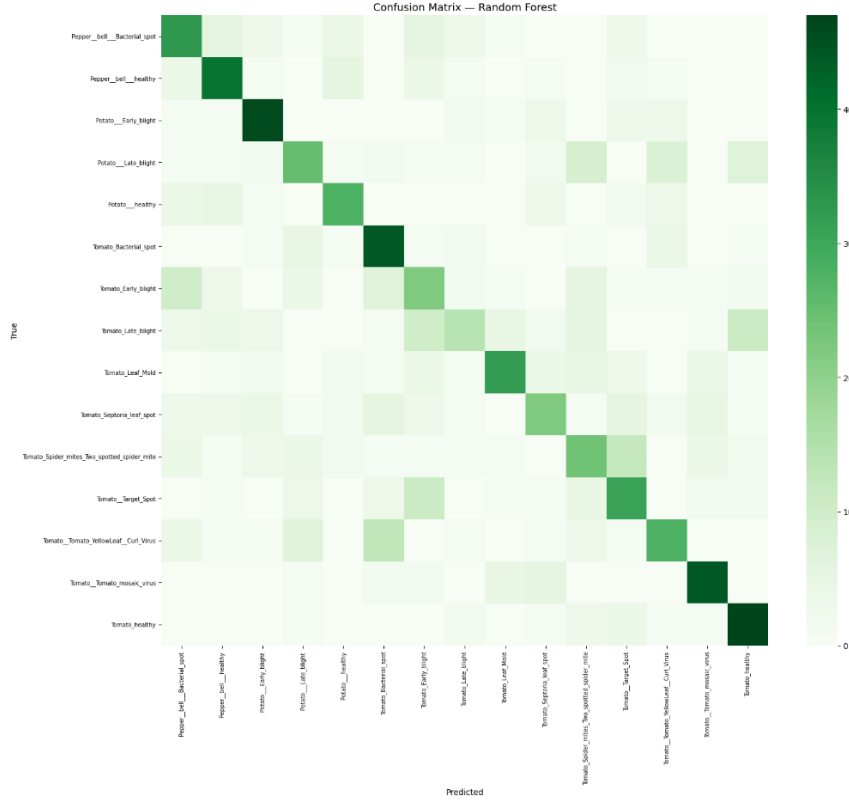


Figure 5. Confusion matrix of the Random Forest model (best classical model).

Despite these promising results, several limitations must be acknowledged. The model's performance may decrease when applied to images captured in uncontrolled environments, particularly those with complex backgrounds or multiple overlapping leaves. Furthermore, the system relies solely on visual features and does not incorporate additional contextual information such as environmental conditions or weather data, which may be relevant for certain diseases. Another limitation is the potential dataset bias, as the training data primarily consists of images collected under controlled conditions, which may not fully represent real-world variability. These findings are consistent with existing literature, where studies have shown that combining multiple data modalities and applying advanced preprocessing techniques can enhance model robustness and generalization capabilities.

VI. Conclusion and Future Work

This work demonstrates that deep learning models, particularly MobileNetV2 with transfer learning, provide an effective solution for plant disease classification. The proposed LeafScan system achieves high accuracy while maintaining computational efficiency, making it suitable for deployment in real-time applications. By integrating a robust deep learning model with an accessible web-based interface, the system bridges the gap between advanced artificial intelligence techniques and practical agricultural use. The system has the potential to support precision agriculture by enabling early detection of plant diseases and facilitating timely intervention. This can contribute to improved crop management and reduced agricultural losses. Future work will focus on enhancing the robustness and applicability of the system. This includes integrating additional data sources such as weather and spectral information to enable multimodal learning, developing a mobile application to increase accessibility, and expanding the dataset to include a wider range of crops, particularly those relevant to local agricultural contexts in Egypt. Furthermore, real-world field testing with farmers will be conducted to evaluate system performance under practical conditions and gather user feedback for further improvements.

VII. References

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